STANFORD WHITE IV

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EDUCATION

National Science Foundation Graduate Research Fellow (NSF–GRFP) *Aug 2023 - Present*
University of Mississippi – One of 2,000 selected from 15,000+ applicants nationwide

University of Mississippi
Ph.D. Student in Engineering Science (Exp. 2028) | M.S. in Engineering Science (2025) | GPA: 4.0
B.S. in Mechanical Engineering (2023) | GPA: 4.0 | Summa Cum Laude | Undergraduate Class Marshal

SKILLS

Machine Learning & AI: PyTorch, TensorFlow, Scikit-Learn, Computer Vision, Multimodal Foundation Models
Programming: Python, MATLAB, Docker, Git, ROS, Arduino, VBA
Engineering Software: Ansys, Abaqus, SolidWorks, Creo Parametric, AutoCAD, Fusion, Inventor
Domain Expertise: Additive Manufacturing, Robotics, Materials Characterization, High-Speed Imaging

RESEARCH EXPERIENCE

Self-Supervised Foundation Model for Polycrystalline Microstructure Representation *Nov 2025 - Present*
University of Mississippi, Oxford, MS
– Pre-trained a ViT-B foundation model with DINO/iBOT++ self-distillation on 217,000+ electron backscatter diffraction (EBSD) scans, using $\mathrm{SO}(3)$ crystal orientations as input with learned invariance to crystal symmetry
– Achieved a median 8.4x label-efficiency improvement over supervised ResNet-50 and a mean 30% accuracy gain on 19 downstream microstructure characterization tasks over DINOv3, MAE, and ResNet-101

Multi-Objective Process Optimization for Metal Additive Manufacturing *Aug 2025 - Present*
University of Mississippi, Oxford, MS
– Optimized laser powder bed fusion (L-PBF) process parameters to control phase fraction in duplex stainless steel, achieving 50% ferrite / 50% austenite for the first time within 3.5% of target composition
– Demonstrated that multi-objective Bayesian Optimization outperforms design of experiments (DOE) for wire-powder directed energy deposition (WP-DED), identifying 2x more feasible printing conditions across 150 experiments

Next-Frame Prediction of Additive Manufacturing Dynamics with High-Speed Video *Aug 2023 - Present*
University of Mississippi, Oxford, MS
– Designed and built a high-speed imaging apparatus integrating an IDT MotionPro Y4-S2 camera with a microscope reflex lens to capture electrohydrodynamic (EHD) jet printing dynamics at up to 60,000 fps
– Implemented a 3D CNN-Transformer model trained on high-speed videos to predict frame-by-frame fluid dynamics, achieving $R^2 > 0.90$ and replacing computationally expensive CFD simulations with 1000x faster inference

Machine Learning-Optimized Drop-on-Demand Inkjet Biofabrication *Jan 2024 - Present*
University of Mississippi, Oxford, MS
– Trained a LightGBM classifier with differential evolution optimization on 240 experiments, achieving stable Silk Fibroin bioink ejection with a >55% success rate, compared to a baseline 4% using traditional approaches

Physics-Informed Transformer for Catalytic Pyrolysis of Polymer Nanocomposites *Aug 2024 - Dec 2025*
University of Mississippi & Center for Graphene Research & Innovation, Oxford, MS
– Developed a physics-informed transformer-based neural network incorporating Avrami-Erofeev kinetic parameters to predict char mass evolution (TGA curves) during catalytic pyrolysis of polyetherimide/graphite nanocomposites
– Achieved $R^2 > 0.98$ on unseen cobalt catalyst data and successfully extrapolated predictions to previously untested catalyst compositions, demonstrating the model's generalizability beyond the training distribution

Machine Learning for Swarm Robotics *May 2022 - June 2023*
West Virginia University & National Science Foundation, Morgantown, WV
– Developed a machine learning pipeline to decompose and imitate decentralized swarm behaviors, achieving over 90% accuracy in replicating agent-level policy weights, communication laws, and motion constraints

WORK EXPERIENCE

NSF Graduate Research Fellowship – Machine Learning Guided Additive Manufacturing *Aug 2023 - Present*
University of Mississippi & National Science Foundation, Oxford, MS
– Awarded the NSF Graduate Research Fellowship, one of approximately 2,000 recipients selected from 15,000+ applicants nationwide, in support of machine learning–guided additive manufacturing research

Undergraduate Research Program Mentor *Aug 2021 - Present*
UM Department of Mechanical Engineering, Oxford, MS
– Spearheaded social media campaigns, photography, and outreach efforts that significantly boosted the visibility and growth of the Undergraduate Research group at UM, recruiting over 150 undergraduate researchers over 4 years
– Led over 100 tours, training events, and presentations—including to U.S. Senator Roger Wicker and an Under Secretary of Defense—that helped secure federal funding for graphene and composite materials research

Vehicle Calibration Intern *May 2023 - Aug 2023*
PACCAR Inc., PACCAR Technical Center, Mount Vernon, WA
– Integrated vehicle software for Peterbilt, Kenworth, and DAF electric and diesel platforms, and conducted testing, calibration, and diagnostics across 460 test cases in support of a fleet of over 300 vehicles
– Developed a MATLAB script to embed 1200+ variable descriptions in calibration files for the vehicle electronic control unit (VECU), enhancing engineers' ability to identify variable functions

Research Experience for Undergraduates – Machine Learning for Swarm Robotics *May 2022 - June 2023*
West Virginia University & National Science Foundation, Morgantown, WV
– Presented research results to subject-matter experts at the West Virginia University Undergraduate Research Symposium and at the American Control Conference, San Diego 2023

PROJECTS

High-Throughput Split-Hopkinson Pressure Bar (SHPB) Data Analysis Software *Jan 2023 - Present*
Personal Project – Targeting Patent Submission
– Built and deployed Python software that automates post-processing for high strain-rate compression testing, increasing throughput 13x (6 to 80 samples/hour)

Mattress Assembly Tolerance Maintainer *Nov 2022 - May 2023*
Senior Design Project
– Designed and prototyped a modular Mattress Assembly Tolerance Maintainer for Ashley Furniture, reducing product rejections and driving a cost savings of $112,000 per quarter

Solero Solenoid Material Selection Study *Aug 2022*
Solero Technologies & UM Undergraduate Research
– Experimentally identified a cost-effective replacement polymer for automotive transmission solenoid covers during a material shortage, achieving 30% greater durability and preventing losses of up to $85,000 per day

PUBLICATIONS

White, Stoddard, et al. "Investigating Nozzle-Based Additive Manufacturing Dynamics with High-Speed Imaging and Machine Learning" *Progress in Additive Manufacturing* (IF: 5.2), Sep. 2026.

White, Salvati, et al. "Physics-informed Machine Learning Prediction of Char Mass Evolution in the Catalytic Pyrolysis of Polyetherimide/Graphite Nanocomposites" *Chemical Physics Impact* (IF: 5.3), Dec 2025.

White, Yang, et al. "Multi-Objective Bayesian Optimization for Geometric Quality Control in Laser Wire-Powder Directed Energy Deposition of 316L Stainless Steel" *In Preparation*, Oct. 2026. (Ready to Submit)

White, Herring, et al. "A Universal Self-Supervised Representation Framework for Polycrystalline Matter" *In Preparation*, Oct. 2026. (Ready to Submit)

White, Tobah, et al. "Machine learning Guided Determination of Processing-Microstructure Relationship in Laser Powder Bed Fusion Produced Duplex Stainless Steels" *Additive Manufacturing Letters*, Oct. 2026. (Under Review)

White, Zhang, et al. "Machine Learning–Enabled Pharmaceutical Security Printing for Drug Authentication and Counterfeit Prevention" *In Preparation*, Oct. 2026. (Ready to Submit)

White, Rahman, Crockett, et al. "Imitating Swarm Behaviors by Learning Agent-Level Controllers" *2023 Proceedings of the American Control Conference (ACC)*, San Diego, CA, 2023, pp. 3657-3662.

PRESENTATIONS

Self-Supervised Foundation Models for Dense Crystallographic Representation in Materials Informatics *Exp. November 2026*
White, Field, et al. - MRS Fall Meeting & Exhibit - Boston, MA

Multiobjective Adaptive Design for Bead Geometry Optimization in Wire-Powder Directed Energy Deposition *Exp. November 2026*
White, Yang, et al. - MRS Fall Meeting & Exhibit - Boston, MA

Machine Learning Guided Exploration of Process-Structure-Property Relationships in Laser Powder Bed Fusion of Duplex Stainless Steels *Exp. November 2026*
White, Tobah, et al. - MRS Fall Meeting & Exhibit - Boston, MA

A Self-Supervised Foundation Model for General-Purpose Microstructure Representation from EBSD Data *October 2026*
White, Herring, Choudhury - IMAT Conference and Exposition - Quebec City, QC

Machine Learning Guided Prediction of Printability During Additive Manufacturing *March 2025*
White, Zhang, et al. - TMS Annual Meeting & Exhibition - Las Vegas, NV

Additive Manufacturing Guided with High-Speed Photography and Machine Learning *March 2025*
White, Han, Choudhury - TMS Annual Meeting & Exhibition - Las Vegas, NV

Imitating Swarm Behavior by Learning Agent-Level Controllers *June 2023*
Mussadequr Rahman, White, et al. - IEEE American Control Conference - San Diego, CA

ACTIVITIES

Brazilian Jiu-Jitsu: Brown belt and volunteer instructor at Midsouth LCCT and 662 Jiu Jitsu
Professional Photography: Wedding, event, graduation, landscape, and sports photography